# Statistics Fundamentals for Data Analysis - Study Notes

Original study notes prepared for the DataMind RAG Assistant project.

## Descriptive statistics

Descriptive statistics summarize the main features of a dataset. The mean (average), median (middle value), and mode (most frequent value) describe the center of a distribution, while the range, variance, and standard deviation describe how spread out the values are.

The median is more robust to outliers than the mean: a handful of extreme values can pull the mean far away from where most of the data actually sits, while the median stays near the bulk of the observations.

## Distributions and the normal distribution

A distribution describes how often each value (or range of values) occurs in a dataset. The normal (Gaussian) distribution is a common, symmetric, bell-shaped distribution that appears frequently in natural and social phenomena, and underlies many classical statistical tests.

Skewness describes asymmetry in a distribution: a right-skewed (positively skewed) distribution has a long tail toward higher values, which is common for variables like income or house prices.

## Correlation vs. causation

Correlation measures how strongly two variables move together, usually summarized with a correlation coefficient between -1 and 1. A value near 1 means the variables tend to increase together, near -1 means one tends to increase as the other decreases, and near 0 means little linear relationship.

Correlation does not imply causation: two variables can be correlated because one causes the other, because a third variable causes both, or simply by coincidence. Establishing causation generally requires controlled experiments or careful causal-inference methods, not just an observed correlation.

## Hypothesis testing basics

A hypothesis test evaluates whether an observed effect in sample data is likely to reflect a real effect in the broader population, or could plausibly be due to random chance. The null hypothesis typically represents 'no effect', and the test produces a p-value: the probability of seeing a result at least as extreme as the observed one if the null hypothesis were true.

A small p-value (commonly below 0.05) is often treated as evidence against the null hypothesis, but a p-value threshold is a convention, not a guarantee of practical significance — a statistically significant effect can still be too small to matter in practice.